Pairs Trading

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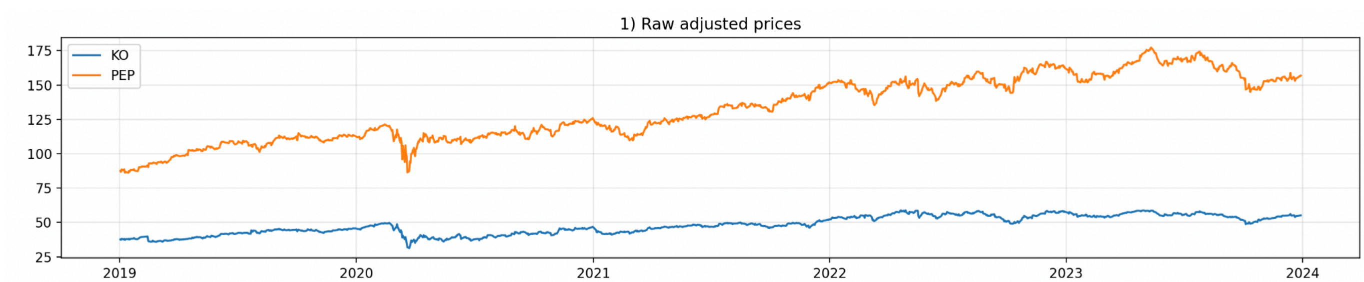
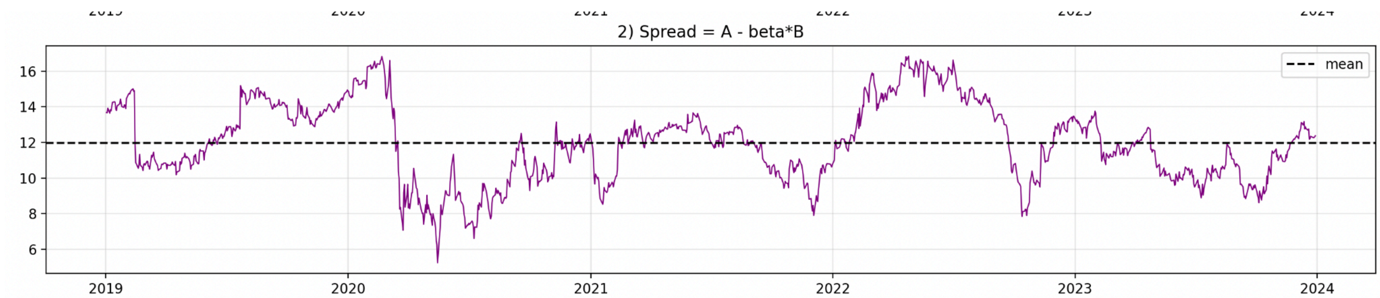
Observations:
For reference, ticker 'a' refers to KO, and ticker 'b' refers to PEP

The cointegration of (a,b) resulted in a p-value of .0711. For the purposes of hypothesis testing, the alpha was set at .05. Any p-value below alpha signifies strong evidence of cointegration between the two observed tickers. However, the observed p-value of .0711 is still a very strong statistical indicator of cointegration between the two observed tickers. Notably, the model was given roughly 4 years of data from mid 2019 to mid 2024, which includes the market data during COVID. An important limitation to note is that 4 years of historical market data may not be a large enough sample to accurately determine the cointegration of two stocks. Sequentially, this could have falsely led us to obtain a p-value closer to statistical significance. However, that could have also falsely led us to obtain a p-value farther from statistical significance. For now, we will operate under the assumption that 4 years of market data is sufficient to determine the cointegration of two stocks.

It should be noted that the engle granger test for cointegration is order dependent. So for testing purposes, cointegration(b,a) was calculated, resulting in a p-value of .0649. Again, Both orders of cointegration resulted in very similar p-values that signify strong statistical significance.





1258 days | train = first 754, test = last 504
Hedge ratio beta (fit on TRAIN only) = 0.254

--- TRAIN (in-sample) ---
Total return : -0.8%
Annualised Sharpe: 0.05
Max drawdown : -21.0%
Number of trades : 20
Win rate : 50.0%

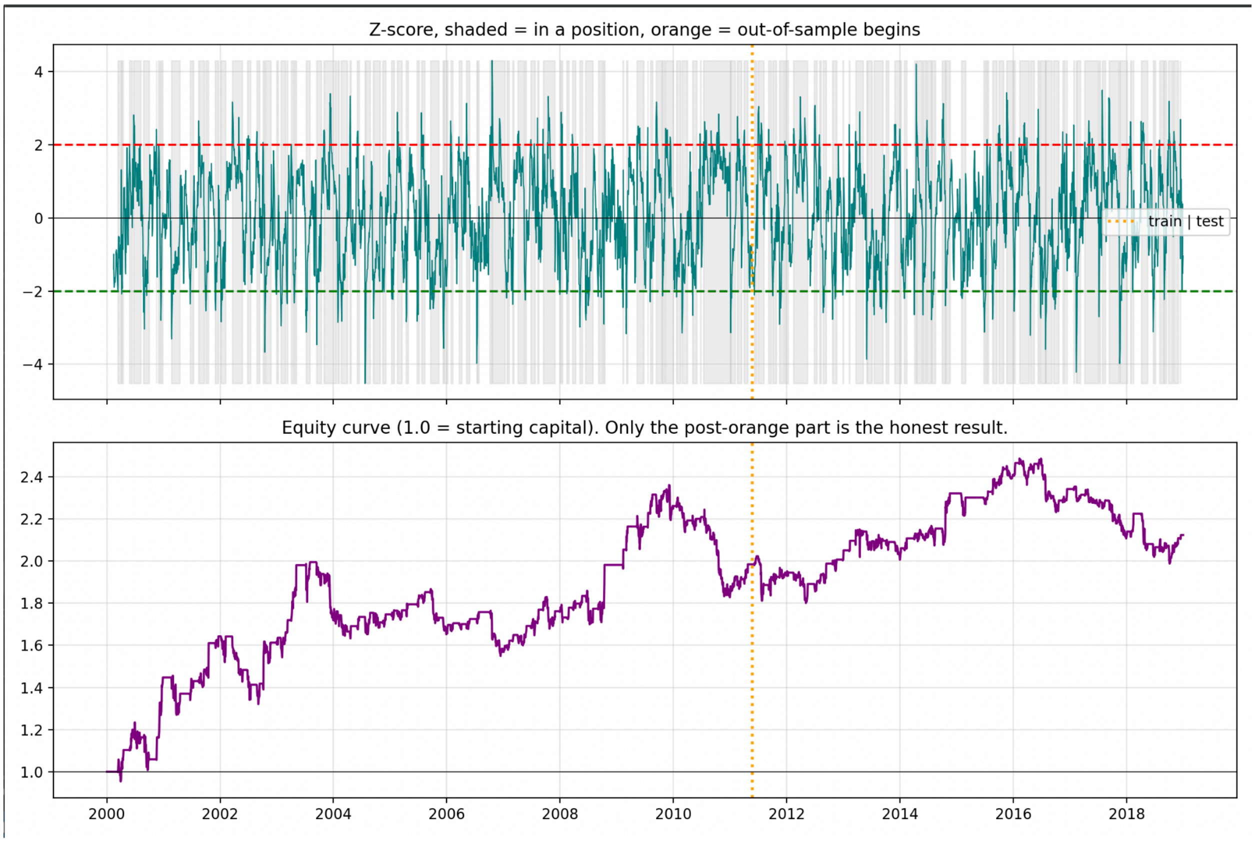
--- TEST (out-of-sample) ---
Total return : -3.1%
Annualised Sharpe: -0.15
Max drawdown : -11.6%
Number of trades : 14
Win rate : 42.9%

The model was using a 60 train and 40 test split on 4 years of market data from 2019 to mid 2024. BPS was set at 1.0 (roughly .01%)

--- TRAIN (in-sample) ---
Total return : 98.4%
Annualised Sharpe: 0.47
Max drawdown : -22.6%
Number of trades : 81
Win rate : 51.9%

--- TEST (out-of-sample) ---
Total return : 7.0%
Annualised Sharpe: 0.15
Max drawdown : -20.1%
Number of trades : 61
Win rate : 41.0%

The model was using a 60 train and 40 test split on 19 years of market data from 2000 to 2019. BPS was set at 20.0 (roughly .2%)



The curve after the orange line is the test data.

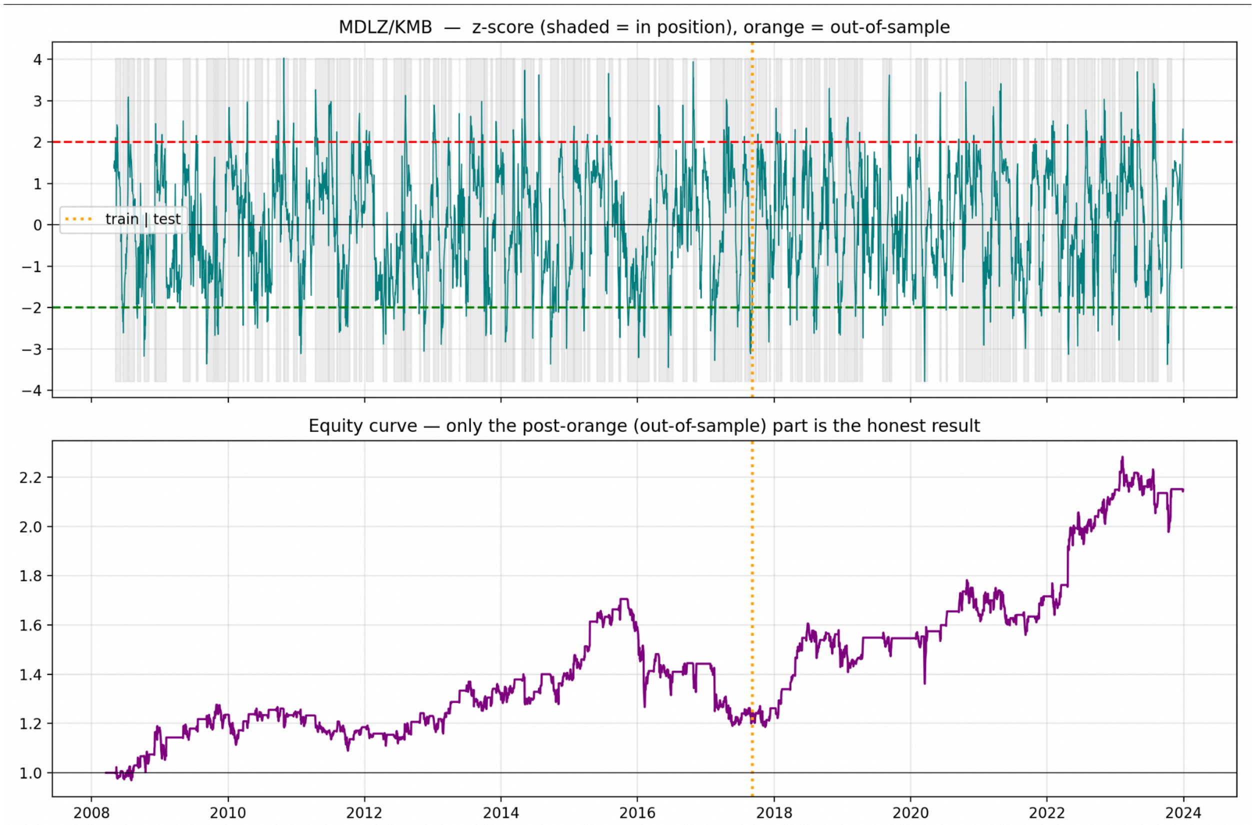
Unfortunately, KO and PEP do not seem to be cointegrated enough for the pairs trading method to yield any significant profits, despite my previous hypothesis. However, that is not to say that these pairs are not cointegrated at all. The backtesting yielded a profit of 34.8% on the test data on the pair, which was quickly offset to about 7% when bps was set to 20.

After testing a few major pairs using a more comprehensive pairs screener, the model screened MDLZ/KMB pairing to be the best, considering the p value, half life, beta and correlation. The half life of the pairing was given the most weight.

Top survivors (ranked by reversion speed):

pair	p_value	half_life	beta	corr
MDLZ/KMB	0.0007	38.0452	0.3794	0.5249
JPM/V	0.0367	62.7972	0.5325	0.5237
BAC/JPM	0.0010	87.1146	0.1695	0.8242
BAC/WFC	0.0048	97.9323	0.1777	0.8307
C/WFC	0.0000	117.0532	-0.1290	0.7188

The pair was then backtested using the pairs trading model. Which yielded the results displayed below



--- TRAIN (in-sample) ---

Total return : 20.6%

Annualised Sharpe: 0.21

Max drawdown : -30.2%

Number of trades : 71

Win rate : 47.9%

--- TEST (out-of-sample) ---

Total return : 78.2%

Annualised Sharpe: 0.72

Max drawdown : -15.2%

Number of trades : 50

Win rate : 48.0%

The results were a little surprising to be completely honest. The model performed very average in its training data, and extremely well in the testing data. This is very counterintuitive. The model performed poorly on a test where it essentially had the "answers", and performed exceptionally when it was completely on its own. The only reasoning that could corroborate this plot would be that the test data was an extremely forgiving time period in the market. The model's win rate stayed at roughly the same 48%, however the profits nearly quadrupled from the initial 20.6% to 78.2% in the test data, indicating that the model "lucked out" its way to larger returns.